C5 past final worksheet

1. (Fall 2018, #10) A deposit of ore contains 100-mg of radium-226, which undergoes radioactive decay. After 500 years, 80.4% of the original mass of radium-226 remains.

   (a) Find the mass $m(t)$ of radium-226 that remains after t years.

   (b) What is the half-life of radium-226?

2. (Fall 2017, #12) The weight of a certain extraterrestrial creature increases at a rate proportional to its weight. A researcher found such a creature at 9am on Monday morning. The researcher weighed the creature at 9am on Tuesday morning, and found it weighed 50g. The researcher weighed the creature again at 9pm on Thursday evening, and found it weighed 150g.

   How much did the creature weigh when the researcher found it? Express your answer as a single fraction, without any exponentials or logarithms.

3. (Fall 2016, #10) The temperature of a glass is measured to be 27° C at 9 pm and 26° C an hour later. The glass is in an air-conditioned room where the temperature is held constant at 20° C. The glass was removed from the dishwasher at a temperature of 37° C. You may use the fact that the temperature T (in degrees Celsius) will obey the equation

$$R(t) = A + Be^{-kt},$$

where t represents the number of hours after 9 pm.

(a) Using the fact that in the long run, the temperature of the glass approaches the temperature of the room, what is $\lim_{t \to \infty} T(t)$? Use this and the information given to solve for all the constants A, B, and k.

(b) When was the glass removed from the dishwasher?

4. (Fall 2015, #9) The number of bacteria in a culture grows at a rate proportional to the total number of bacteria. The starting population contains 200 cells. After four hours, there are 1000 cells.

   (a) Find the expression for the number of bacteria after t hours.

   (b) How much time will pass before the population reaches 5000 cells?

5. (Spring 2015, #8) The half-life of Ca-14 is 5730 years.

(a) Find a formula for the amount of Ca-14 present after t years in a sample that initially contains A grams.

(b) How long will it take for the sample to degrade so that it contains only one third of its initial amount?

(c) If the initial sample contains 100 grams of Ca-14 how long will it take until there are only 10 grams left?